DIY GREENHOUSE BUILD

Build Your Own Temperature & Humidity Gauge

A beginner's guide: no coding and no experience needed.

This little gauge sits by your plants and shows the **temperature** (°F) and **humidity** (%) on a small screen, updated every 10 seconds. If you want, it can also join your home Wi-Fi so you can check it from your phone.

Plug and play: the software is already written. You don't write or change any code: you plug the board into a computer, open **chetsemail26.github.io/diy-greenhouse-gauge** and click **Install**.

The whole build in 3 steps

1

Get the parts

3 small electronic boards plus a few wires and a USB cable. (Page 2)

2

Connect the wires

8 wires, all labeled. No electronics knowledge needed. (Page 3)

3

Install the software

One button in your web browser. About 2 minutes. (Page 4)

How it works (the 30-second version)

Part	Its job
D1 Mini (the "brain")	A tiny computer the size of a postage stamp. It reads the sensor, draws on the screen and talks to Wi-Fi.
SHT31 (the sensor)	Measures temperature and humidity. It's accurate: about ±0.5°F and ±2% humidity.
OLED screen	The 1.3-inch display that shows the numbers.
USB cable	Powers everything, and is used once to install the software.

What you need to know before starting

Time: about 30-60 minutes. **Cost:** a few small parts, often sold in multi-packs. **Computer:** any Windows, Mac, Linux or Chromebook computer with the **Chrome** or **Edge** browser (only needed for setup). **Soldering:** these boards often come with loose pins that need soldering on. If you'd rather skip soldering, look for boards sold **"with headers soldered"** or "pre-soldered."

STEP 1

Get the parts

Search for these names on Amazon, AliExpress or any electronics shop. The search terms in the right column find the right thing.

Part	Qty	What to look for	Search for
D1 Mini board	1	ESP8266 based. Most come with a "CH340" USB chip; that's fine.	`Wemos D1 Mini ESP8266`
SHT31 sensor	1	A small blue or purple board with 4-6 pins. A **probe version** (sensor sealed in a mesh tip on a cable) is great for humid greenhouses.	`SHT31 sensor module`
1.3" OLED screen	1	**Must be the 4-pin "I2C" version** (pins: VCC, GND, SCL, SDA). The 6-7 pin "SPI" version won't work.	`1.3 inch OLED SH1106 I2C 4 pin`
Jumper wires	~10	Female-to-female if your boards have pins sticking out.	`dupont jumper wires female`
Mini breadboard	1	Optional but easiest: it lets two wires share one pin (see page 3).	`mini breadboard 170`
USB cable	1	Must fit your D1 Mini (usually micro-USB) and must be a **data** cable, not "charge only."	`micro USB data cable`
USB power adapter	1	Any phone charger. The gauge uses very little power.	(the one in your drawer)

> **The #1 beginner trap:** many USB cables only carry power, not data. If your computer doesn't see the board in step 3, swap the cable before anything else.

Nice to have

A 4-wire cable about 1 m (3 ft) long, so the sensor can sit inside the greenhouse while the board and screen stay outside, away from moisture. Keep the sensor cable to about 1 m; longer runs can make readings fail.

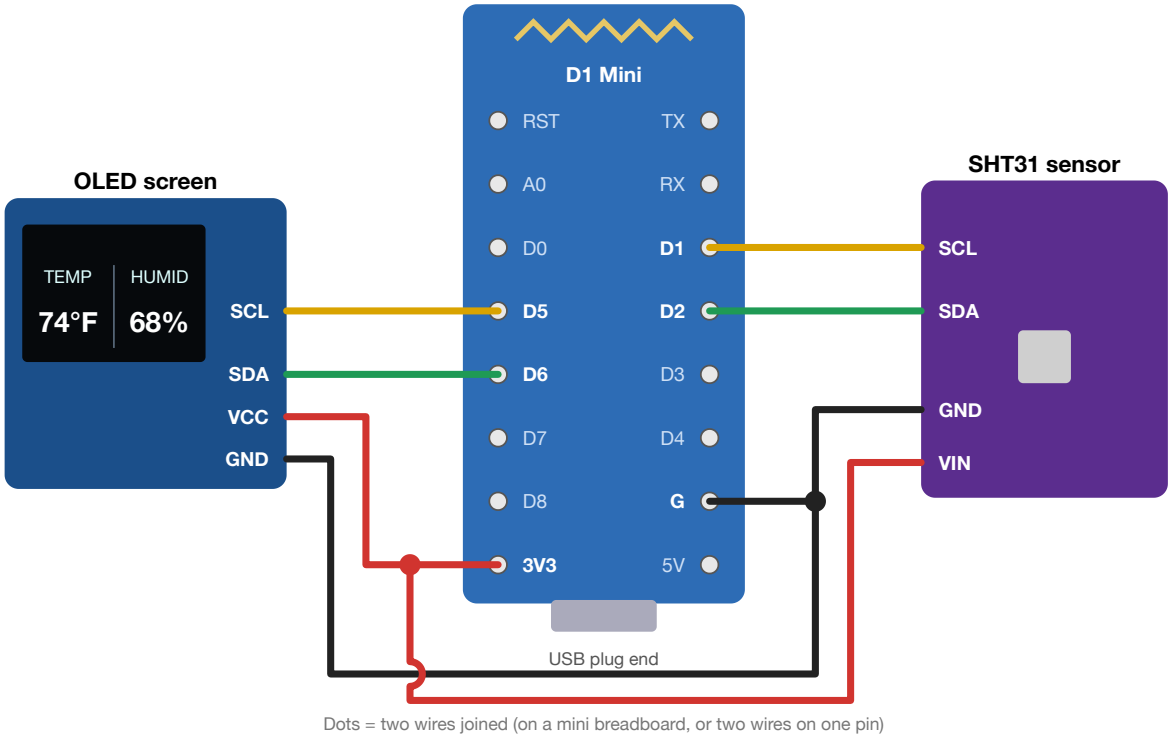
Tools

A soldering iron (only if your boards came with loose pins). Nothing else.

STEP 2

Connect the wires

Every pin has a tiny printed label. You're just matching labels. **Keep the USB cable unplugged while you wire.**



Dots = two wires joined (on a mini breadboard, or two wires on one pin)

Wire	From the SCREEN pin	To D1 Mini	From the SENSOR pin	To D1 Mini
power	**VCC**	**3V3**	**VIN** (or VCC)	**3V3**
ground	**GND**	**G**	**GND**	**G**
clock	**SCL**	**D5**	**SCL**	**D1**
data	**SDA**	**D6**	**SDA**	**D2**

Two parts, one power pin

The screen and the sensor both need **3V3** and **G**, but the D1 Mini has only one of each. Easiest fix: plug the D1 Mini into a mini breadboard and push both wires into the same row as the pin. (Wires in the same row are connected.) A small splitter or two wires soldered to one pin also works.

Double-check these:

- Use **3V3**, not 5V.
- Pin *order* differs between brands, so always go by the **printed labels**, not the position. Screens often swap VCC and GND.
- Sensor has extra pins (ADR, AL)? Leave them empty.

STEP 3

Install the software

One button in your web browser. Nothing to download, nothing installed on your computer.

1. Plug the D1 Mini into a computer with the USB **data** cable.
2. On that computer, open **Chrome** or **Edge** and go to:
   chetsemail26.github.io/diy-greenhouse-gauge
3. Click **Install**. Pick the board from the list (it's called something like *USB Serial*, *CH340* or *usbserial*) and click **Connect**. Wait until it says it's done (about 1-2 minutes). Don't unplug it during this.
4. **Wi-Fi (optional):** when it asks, pick your home Wi-Fi and type its password. Or skip it: the gauge works fine without Wi-Fi.

That's it. The screen shows **TEMP** and **HUMIDITY**. From now on, power it from any phone charger.

> **The board doesn't show up in the list?** 1) Swap the USB cable; it's most likely a charge-only one. 2) Try another USB port. 3) On Windows, or on an older Mac, install the free "CH340 driver" from the chip maker, WCH (search `CH340 driver WCH`), then try again. 4) Safari and Firefox can't do this. Use Chrome or Edge.

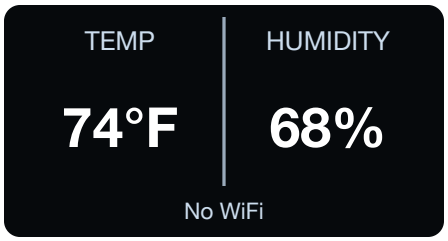
If you added Wi-Fi

On a phone or computer **on the same Wi-Fi**, open `greenhouse-gauge.local` to see live readings. If that doesn't open (common on Android), find the gauge in your router's app under connected devices and open its number address (like `192.168.1.57`).

Add or change Wi-Fi later, from your phone: whenever the gauge isn't on Wi-Fi, it makes its own network called **Greenhouse-Gauge**. Join it from your phone, and a setup page pops up (if it doesn't, open `192.168.4.1`). Pick your Wi-Fi, type the password, tap **Save**.

Home Assistant users: the gauge shows up on its own under Settings → Devices & services.

Reading the gauge



On the screen	Meaning
74°F / 68%	Current reading, refreshed every few seconds.
--	No reading from the sensor right now (see the blue light below).
No WiFi	Not on Wi-Fi. Normal if you skipped Wi-Fi.
SENSOR FAIL	The sensor stopped answering and the gauge couldn't fix it on its own.

The little blue light on the D1 Mini

Blue light	Meaning	What to do
Off	Everything is working.	Nothing.
Slow blink (1 second on, 1 off)	No reading for 30+ seconds. The gauge is trying to fix itself.	Wait a few minutes. It usually recovers on its own.
Fast blink	It gave up after 5 tries ("SENSOR FAIL").	Check the sensor wires, then unplug and replug the gauge.

It fixes itself. If the sensor goes quiet for 2 minutes (a wiggly wire, a power blip), the gauge restarts itself, up to 5 times. When the sensor comes back, everything resets. The screen has its own wiring, so it keeps working even when the sensor doesn't.

Where to put it

Tip	Why
Keep the D1 Mini and screen **outside** the humid area. Only the sensor goes in.	Moisture and condensation can damage the boards.
Put the sensor at **plant height**, in the shade.	In direct light it measures the lamp, not the air your plants feel.
Keep it **away from heat mats** and heaters.	Same reason: you want the air temperature.
Keep the sensor a few inches from the D1 Mini.	The board makes a little heat, which reads warmer and drier.
Building a case? Leave **air holes** by the sensor and a window for the screen.	A sealed sensor can't measure the room.

If something's not right

Problem	Fix
Computer doesn't see the board	Swap the USB cable (most likely a charge-only one), try another port, install the CH340 driver (page 4). Use Chrome or Edge.
Screen stays black	Check the screen's 4 wires: **SCL** → **D5, SDA** → **D6**. Check VCC/GND aren't swapped (read the labels). Make sure it's the 4-pin I2C screen.
Screen works but shows **--**	Check the sensor wires: **SCL** → **D1, SDA** → **D2**, VIN → 3V3, GND → G. Then unplug and replug the gauge.
Readings jump around when you touch it	Loose wire. Re-seat the jumper wires, or solder the joints.
Reads too warm or too dry	The sensor is near the board, a lamp or a heat mat. Move it (page 5). Your hand warms it too, for a few minutes.
Stuck near 100% humidity after misting	Water on the sensor. It recovers as it dries. Mist away from it.
Can't find the **Greenhouse-Gauge** Wi-Fi	It only appears while the gauge isn't on Wi-Fi. Unplug it, plug it back in and wait 1 minute.
`greenhouse-gauge.local` won't open	Make sure your phone is on the same Wi-Fi, or use the gauge's number address from your router's app.
Wi-Fi won't connect	The gauge only joins 2.4 GHz networks. Check the password; it's case sensitive.

Word list

D1 Mini / ESP8266	The tiny Wi-Fi computer board. ESP8266 is the chip on it.
Firmware	The software that lives on the board.
Flashing	Installing firmware onto the board (step 3).
ESPHome	The free, open-source system the firmware is built with.
I2C	The simple 2-wire language (SCL = clock, SDA = data) the sensor and screen use to talk to the board.
3V3 / G	3.3 volts (power) and Ground. Every part needs both.

For tinkerers: the full settings file, `greenhouse-gauge.yaml`, is at github.com/chetsemail26/diy-greenhouse-gauge. It's built with ESPHome (esphome.io), so you can change the layout, switch to °C, or add features. Plug-and-play users can ignore it.